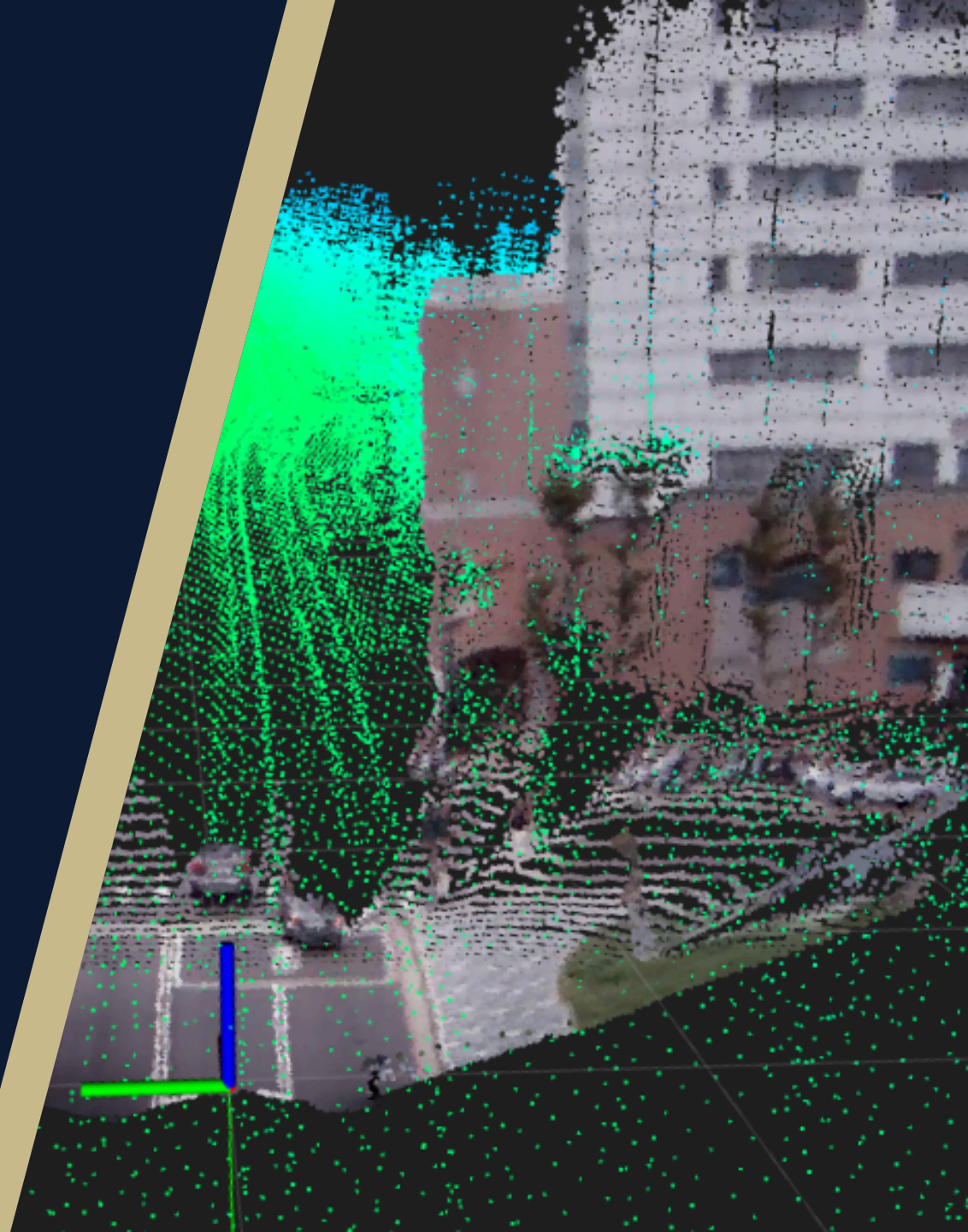
SDC 2026 · MIDTERM 01

Localizing a Camera on a LiDAR Map

NDT scan matching + EKF, one pipeline for three tracks

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The task: place a camera on a LiDAR map

In: point clouds from camera images, no absolute scale

Against: a pre-built LiDAR map

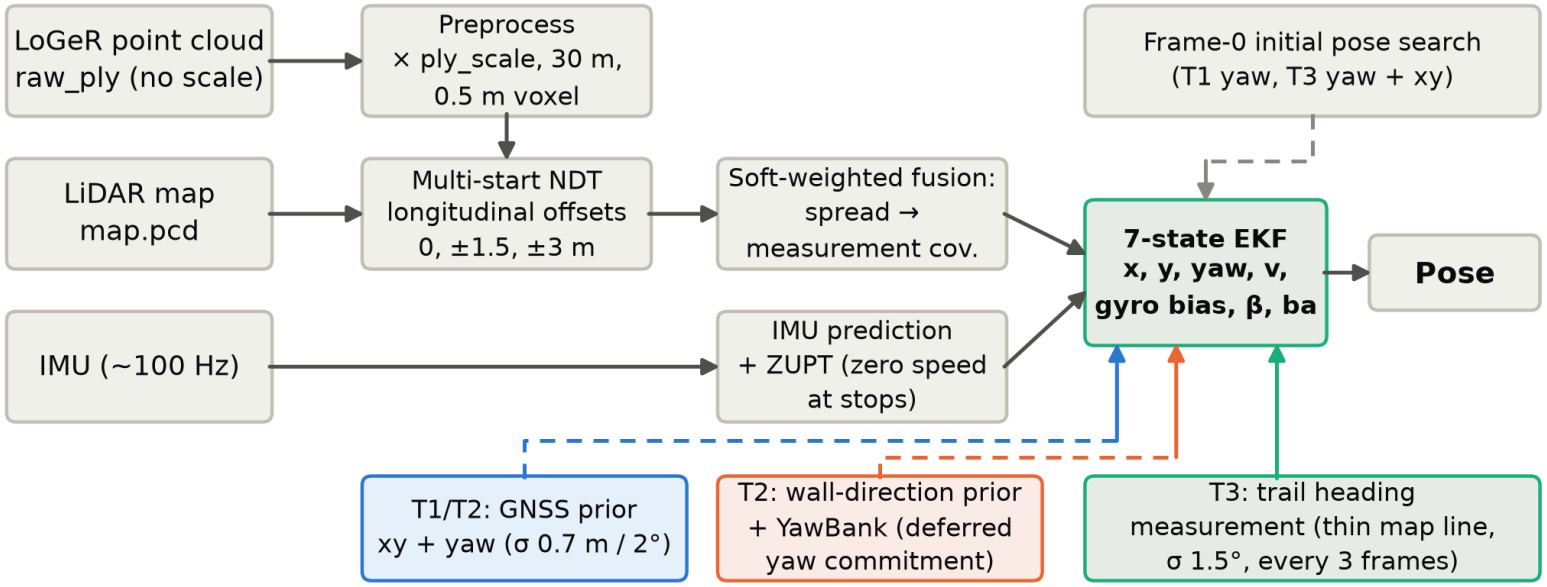
Score: E = RMSE(position, m) + 0.2 · RMSE(yaw, deg)

Track	Weight	Baseline	Ours
1	0.3	1.79	**1.13**
2	0.3	1.92	**0.77**
3	0.4	1.29	**1.74**

Kaggle public scores, lower is better. Track 3 private baseline: 1.40.

One pipeline, three tracks

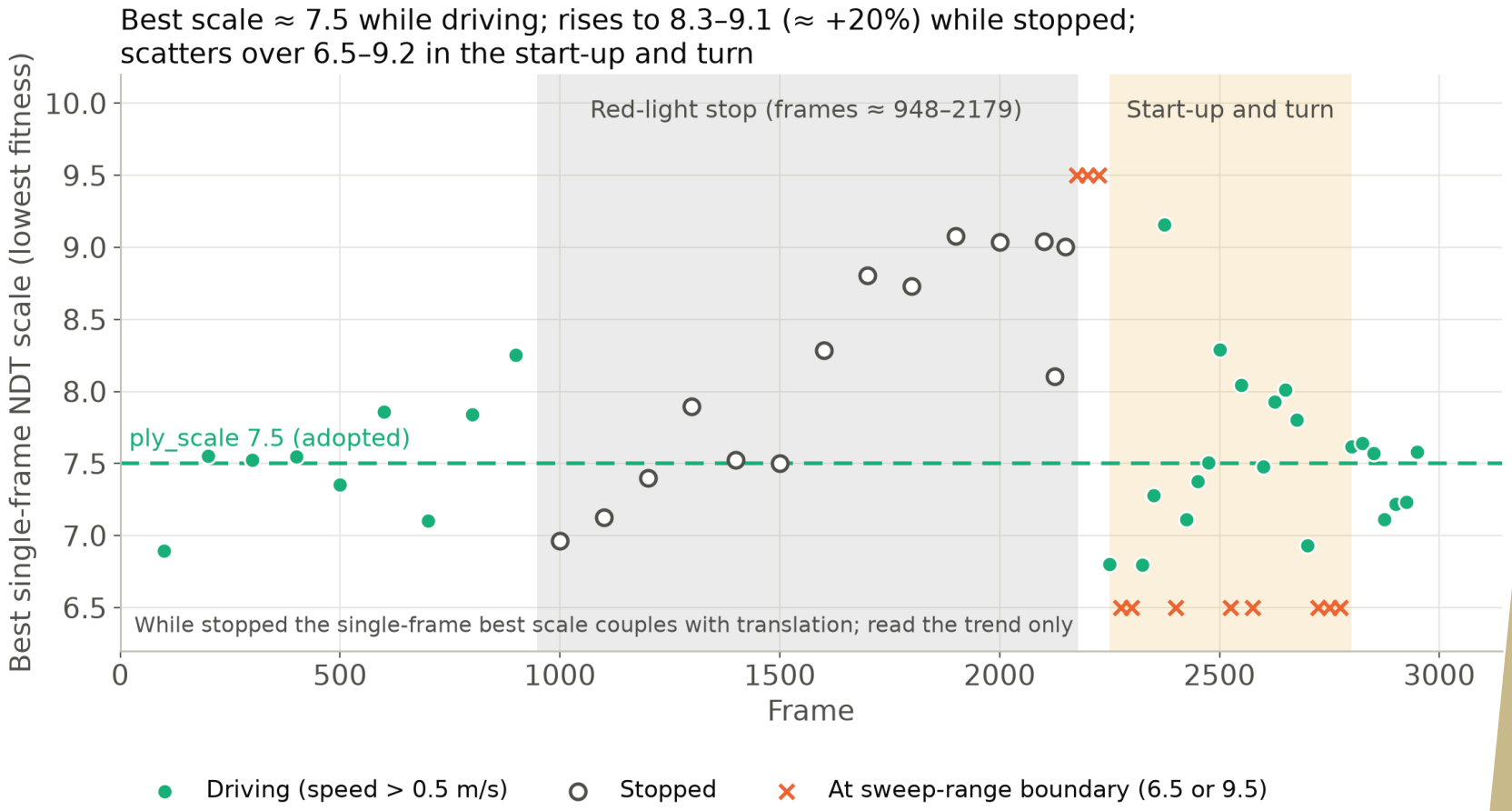
One pipeline; each track enables its own add-on modules via yaml (dashed = enabled on that track only)



KEY POINTS

- NDT proposes, the EKF decides
- Add-ons switched per track in yaml
- Causal: no future frames

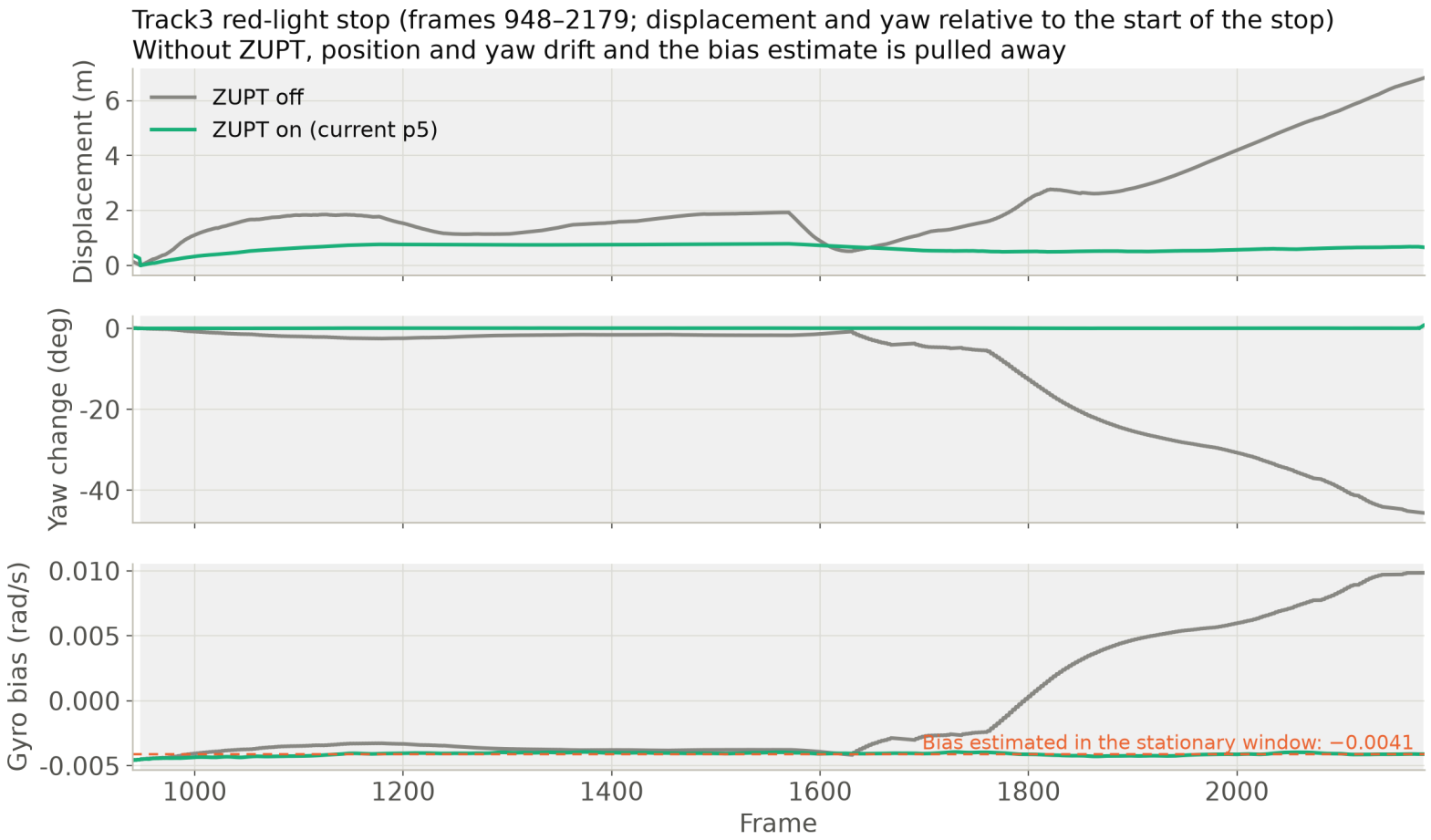
Point-cloud scale



DETAILS

- Single-frame scale sweeps pick ply_scale
- T1 8.5 · T2 8.7 · T3 7.5
- When stopped, best scale couples with translation

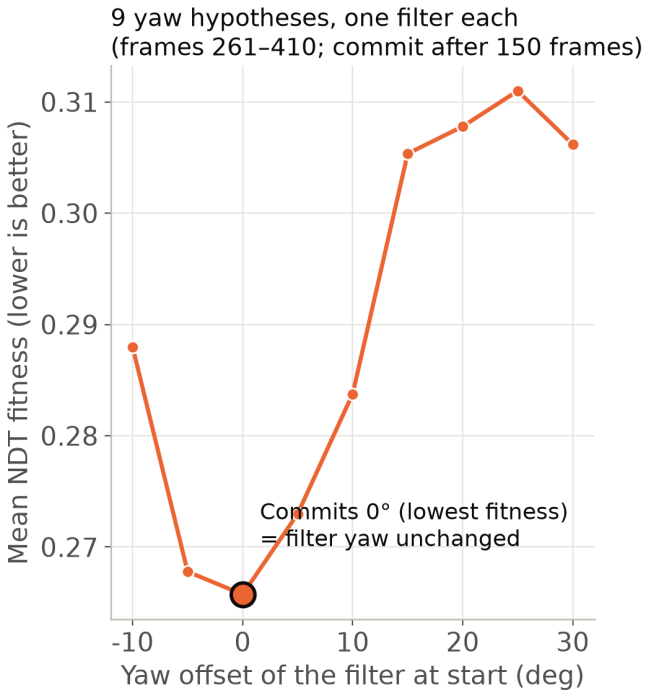
ZUPT at a red light

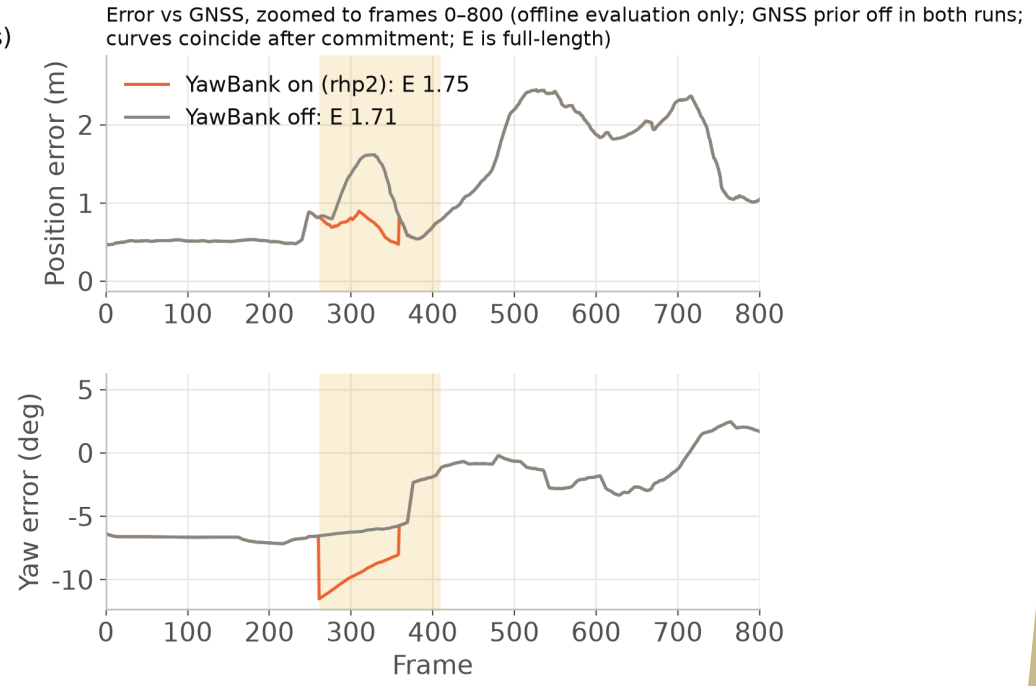


DETAILS

- An IMU window detects standstill
- Pins speed and gyro bias
- Stop drift: 6.8 m → 0.7 m, yaw −46° → +0.7°

Track 2: deferring the yaw decision



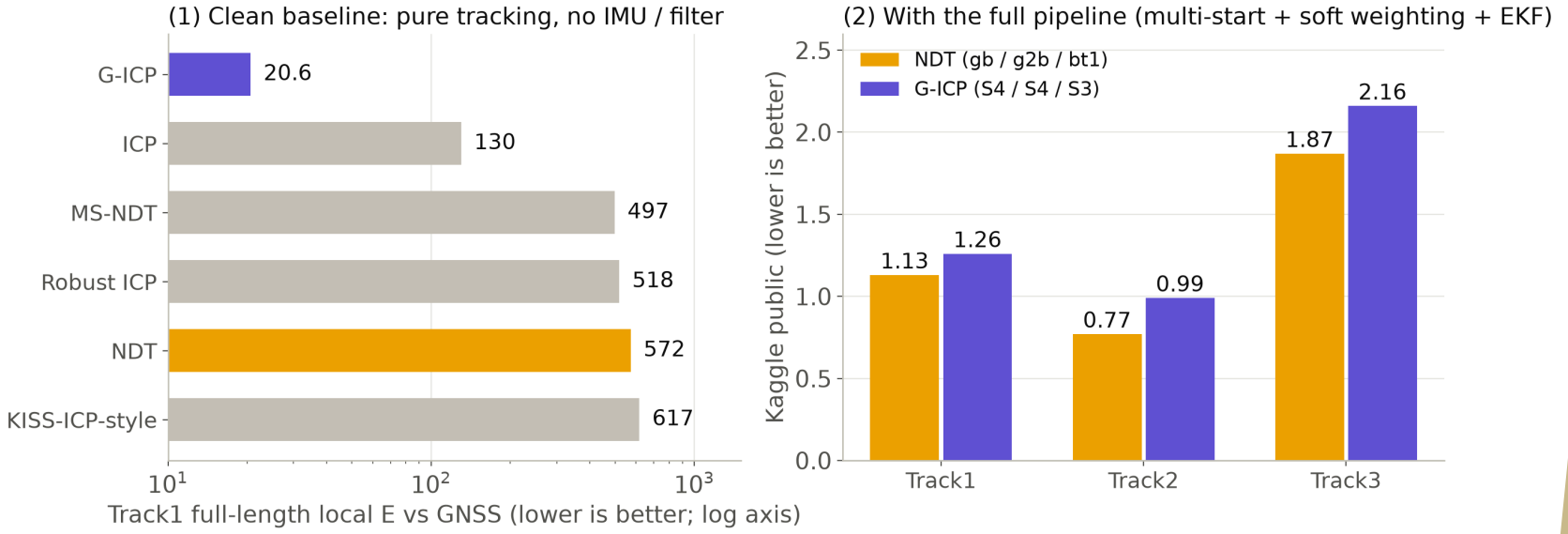


DETAILS

- 9 yaw hypotheses, commit after 150 frames
- It commits 0° with the wall-direction prior
- Bank on / off: E 1.75 / 1.71 (GNSS off)

Why NDT and not G-ICP?

Bare G-ICP is the most accurate, but NDT wins on all three tracks once the full pipeline is added



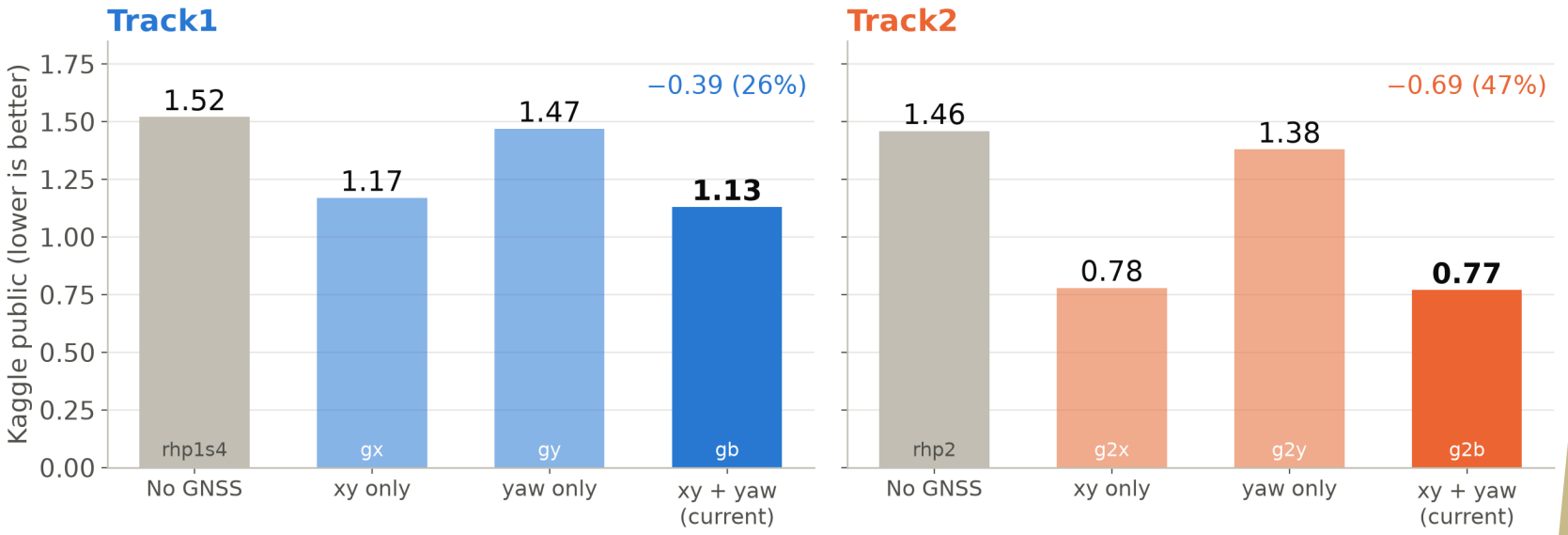
Left: Track1 only. Right: the NDT reference for Track3 is bt1 (1.87); current p5 is 1.74.

KEY POINTS

- Bare G-ICP drifts least
- In the full pipeline NDT wins on all 3 tracks
- G-ICP has a systematic longitudinal bias

GNSS prior on Tracks 1 and 2

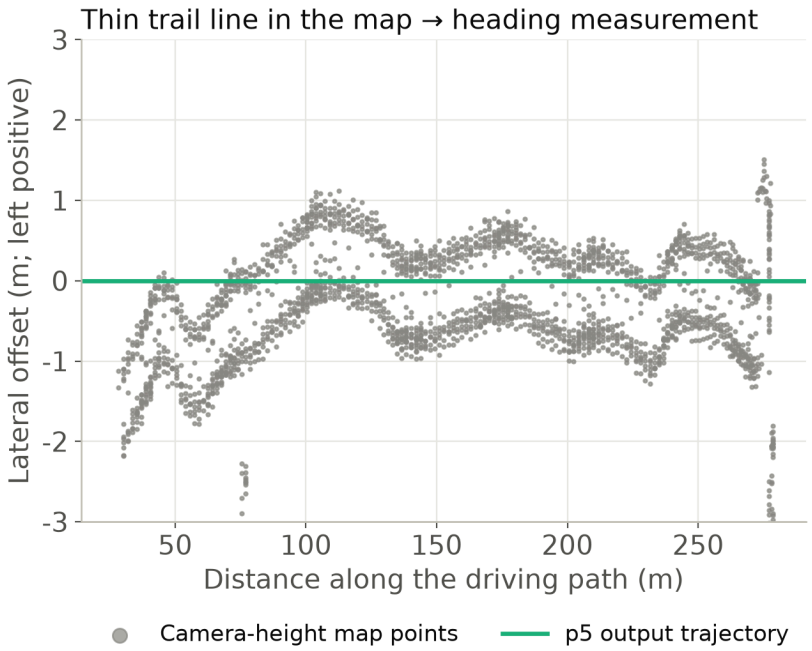
Contribution of the GNSS prior (σxy 0.7 m, σyaw 2°)
GNSS use is allowed by the official documents

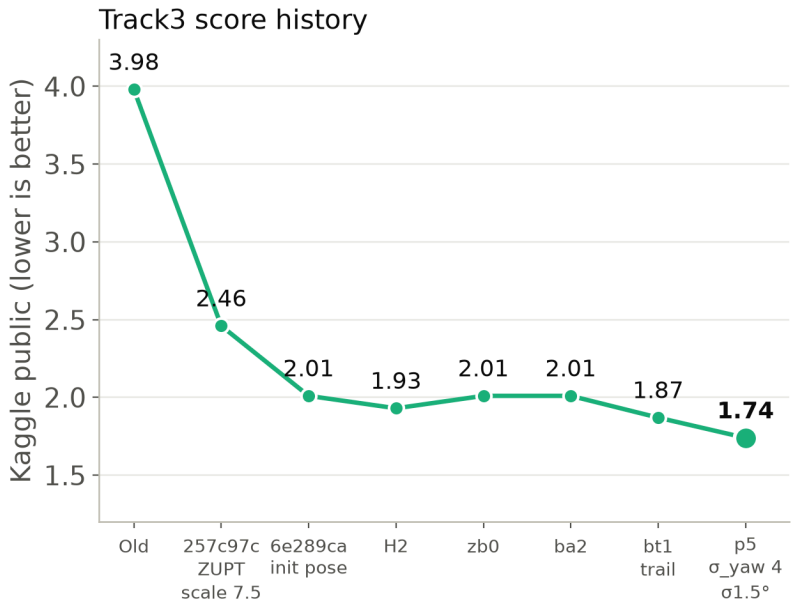


KEY POINTS

- Allowed by the official documents
- Causal, about 4 Hz, σ 0.7 m and 2°
- Track 1: −0.39, Track 2: −0.69

Track 3: a heading cue from the map



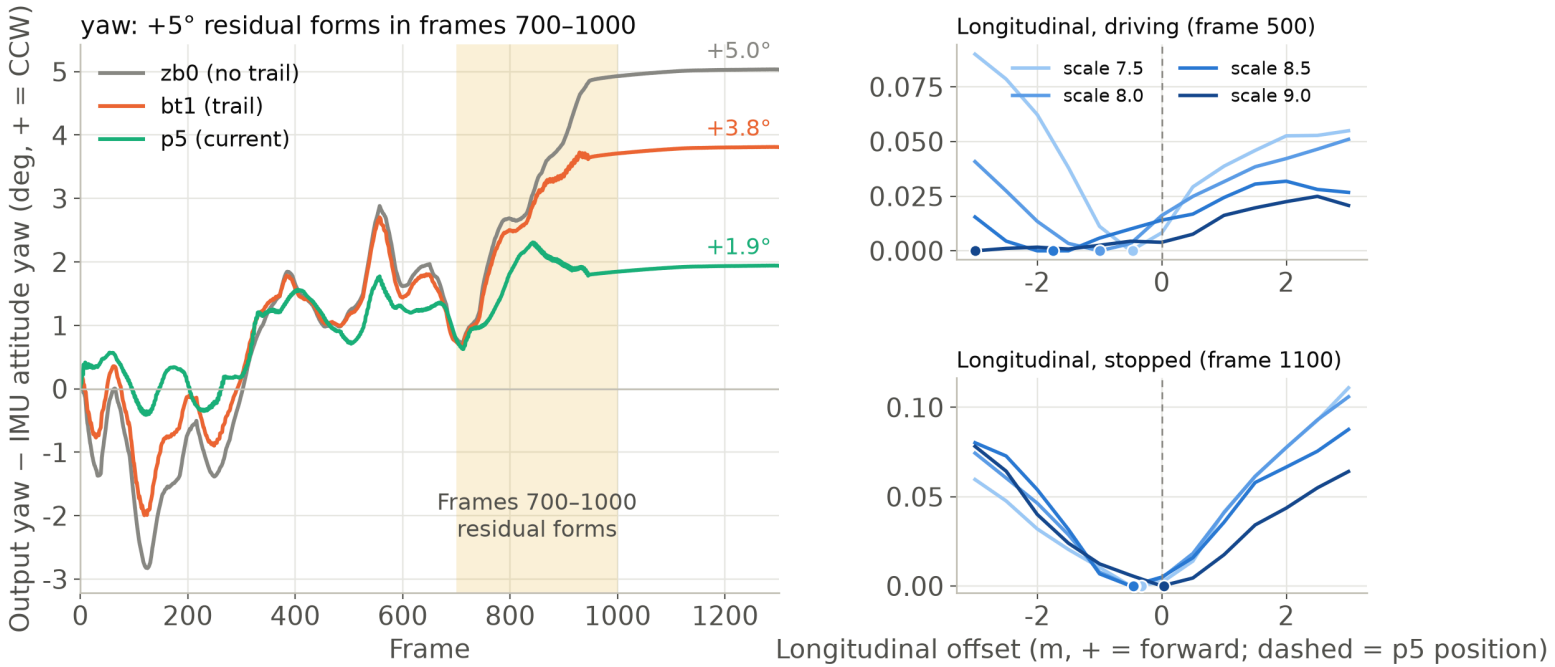


KEY POINTS

- A thin point line in the map gives the road direction
- Heading measurement, σ 1.5°, every 3 frames
- Kaggle 3.98 → 1.74

Where the Track 3 error lives

Lateral and yaw errors are structured and fixable;
longitudinal is degenerate: its best position moves with scale (≤0.5 m offset while stopped)



KEY POINTS

- Lateral and yaw: structured, fixable
- Longitudinal: degenerate, coupled to scale
- Stopped offset ≤ 0.5 m

What we tried and dropped

Idea	Result
Covariance floor	T1 1.56 → 1.62; T2 diverged
G-ICP matcher	Worse on all 3 tracks
Trail update at stops	T3 1.87 → 1.92
Lateral nudge	T3 1.74 → 1.84
Ground z constraint	Rejected by visual check

RULE

- Keep only what Kaggle or a visual check supports

Takeaways

- One NDT + EKF pipeline; each track switches on its own add-ons
- Kaggle public: Track 1 1.13 · Track 2 0.77 · Track 3 1.74
- Track 3: remaining error is longitudinal (scale-coupled) and yaw

Thank you · Q&A